

izi — XAU Assistant: operator manual

Generated 2026-08-26 by scripts/build_manual.py. Get this in Telegram any time with /manual.

What this is

An MT5 trading assistant for XAUUSD in two halves. The MQL5 Expert Advisor (runs inside MetaTrader 5 on Windows) holds the strategy and is the sole decision maker: it evaluates every closed bar, executes in AUTO mode, and manages the basket. The Python service (runs in WSL2, port 9000) grades each signal with an AI forecaster, sends every alert to Telegram, and logs everything to SQLite.

Two rules explain most behavior. (1) The strategy decides; the AI only grades. In AUTO the EA executes FIRST, then asks the AI. (2) Fail-open everywhere: if the AI service is down, strategy signals still alert and execute, marked 'AI unavailable'. A dead service never blocks a trade.

The moving parts

MT5 (Windows): XauAssistant EA on a XAUUSD chart

Service (WSL2): FastAPI on `http://127.0.0.1:9000`

Database: `service/xau_assistant.db` (SQLite)

Telegram bot: alerts + remote control (below)

The EA heartbeats the service every 5 seconds; remote commands (manual trades, exits, stop moves) ride back on the next heartbeat, so a tap takes at most ~5 s to reach MT5.

New client setup

One script does almost everything and is safe to re-run at any time — finished phases skip:

```
cd <repo> && scripts/setup.sh
```

It creates the venv, installs dependencies, runs the tests, starts the service, links Telegram (message your bot once when prompted), copies the EA into the MT5 data folder and compiles it.

Configuration

```
cd service && cp .env.example .env
```

FORECASTER=fake runs without the AI model (deterministic stand-in) — useful on machines without torch. Leave the default for real grading.

Two manual MT5 steps (MT5 stores these encrypted)

1. Tools > Options > Expert Advisors: tick 'Allow WebRequest for listed URL' and add exactly:

```
http://127.0.0.1:9000
```

2. Drag XauAssistant (Navigator > Expert Advisors) onto a XAUUSD chart and tick 'Allow Algo Trading' in the dialog. If the EA was already on the chart, remove and re-attach it.

Live accounts

On a real-money account AUTO refuses to trade unless the EA input AllowLiveTrading is set to true. This is deliberate: flip it only when the logged AI accuracy says the system has earned it.

Daily operations

Come up

Double-click xau-launch.bat on the Desktop — it boots MT5, WSL and the service. Or by hand:

```
cd <repo>/service  
nohup .venv/bin/uvicorn app.main:app \  
    --host 127.0.0.1 --port 9000 >> service.log 2>&1 &
```

The first AI forecast after a start can take minutes (model load); everything after is fast.

Trading is not blocked meanwhile (fail-open).

Is it running?

```
curl http://127.0.0.1:9000/health
```

...and in Telegram, /status: the first line shows the EA connection ('EA: connected (Ns ago)'). If MT5 restarted, verify the EA is still on the chart — it can detach silently; re-attaching is manual.

Shutdown

```
pkill -f "uvicorn app.main"    # exit code 144 is normal
```

...then close MetaTrader 5. Risk state (kill switch, drawdown watermark, daily brake) survives restarts — it lives in MT5 global variables, not in memory.

Restart after a code change

Kill first (foreground), start second (background) — never both in one backgrounded command.

The service does not auto-reload. For EA changes, re-run scripts/setup.sh (copies + compiles; expect '0 errors'); the terminal reloads the EA automatically after a successful compile.

Telegram commands

Everything is also pinned in the chat. Commands reply only to the owner chat; a linked broadcast channel sees trade activity but never account figures or buttons.

```
/status    snapshot: session, EA + mini-app connection, protection
/bal       balance, equity, floating P/L, today + week P/L
/mode      execution AUTO/MANUAL, entry ADR/FIXED, lane M5/M15
/agree     confirmation modules: higher-TF + EMA-200 agreement
/trade     manual entry: [BUY] / [SELL] buttons, tap = confirm
/config    current settings incl. entry mode
/chart     live chart (mini app) or a rendered snapshot
/stats     per-strategy signal hit-rates
/history   last 10 trade events
/channel   link/unlink the broadcast channel
/help     the pinned command reference again
/manual    this PDF
```

How a manual trade works

/trade shows the live price and two buttons. Tapping one queues the entry; the EA opens it on the next heartbeat, subject to ALL its own risk gates (kill switch, daily brake, spread, sizing). Once open it is a normal EA basket: managed, alerted, and auto-exited like any other. Guards refuse the tap when a trade is already open, an entry is already queued, or the EA is disconnected.

Buttons and troubleshooting

Buttons you will meet

[Take trade]/[Skip] MANUAL mode entry proposals
[EXIT - close trade] on trade-open notices: close the basket
[EXIT] + [Move SL] on the FIXED-ride target alert
[Reset brake] on daily-loss-brake notices ($\geq 70\%$ spent)

[Move SL to here] ratchets every leg's stop to the current price, locking the gain while the ride continues. It only ever tightens, and the buttons return after each move, so later taps lock more. Old buttons stay tappable forever — every tap is re-checked, so a stale tap degrades to a polite refusal, never a wrong trade.

Quick diagnosis

EA shows disconnected: is MT5 open, the EA on the chart (smiley icon), Algo Trading enabled, and the WebRequest allowlist set? Indicator lines missing after switching timeframes: they repaint on the first tick; if the market is closed they return with the next tick. A trade did not fire: /status — check protection (brake or kill switch), the trading window, and spread; refusals are also alerted with the real reason.

Where things live

service/service.log service log (rotated at 20 MB)
service/xau_assistant.db signals, trades, heartbeats (SQLite)
MT5 'Experts' tab EA-side log
scripts/setup.sh re-run any time; it self-skips